

A Priority-based Semi-Grant-Free NOMA: Protocol Design and Performance Analysis

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Abstract—This paper introduces a novel semi-grant-free non-orthogonal multiple access protocol that ensures reliable massive connectivity of both scheduled and random access devices of diverse data rate (DR) requirements while tripling the network's capacity. In more detail, the protocol classifies the devices into two categories, namely primary and secondary. Primary devices (PDs) require high-data-rate scheduled or random access with high-transmission probability, and are one-third of the total number of devices, while, the rest, are classified as secondary devices (SDs) and have low-DR requirements as well as access the network in a random fashion. The base-station generates a number of radio resource blocks (RRBs) that is equal to the number of PDs and allocates a primary and two SDs in each RRB. The allocation is based on the requested DRs of the primary and SDs as well as on the transmission probability of the SDs. To prove the feasibility and efficiency of the presented protocol, we conduct a performance analysis that results in the extraction of novel and insightful closed-form expressions for the outage probability and achieved throughput for all devices.

Index Terms—Outage probability, performance analysis, semi-grant-free non-orthogonal multiple access, theoretical framework

I. INTRODUCTION

THE emergence of next-generation internet-of-things (NG-IoT) applications, such as the fourth industrial revolution, agriculture, transportation, and healthcare, demand the wireless network to ensure high reliability, low latency, and massive connectivity of devices with diverse data rate (DR) requirements [2]–[5]. This is expected to considerably stress the radio access network and create the so-called spectrum scarcity problem [6]–[9]. As a consequence, the need for access protocols that enhance radio resource utilization has appeared. As a response to this, the non-orthogonal multiple access (NOMA) [10]–[14] and grant-free (GF) NOMA concepts have emerged [15]–[17].

The adoption of NOMA as one of the core enablers of next-generation wireless networks has been the focus of several research works, including [18]–[27]. In particular, the authors of [18] provided a thorough overview of the research challenges associated with the adoption of NOMA in wireless networks and outlined the future research directions. Similarly,

in [19], the authors carried out a comprehensive survey with respect to NOMA schemes. They provided an overview of the main design principles and key principles and discussed the research challenges and future research directions. Finally, Elbayoumi *et al.* presented a thorough overview of the application of NOMA schemes towards supporting massive numbers of connected devices and evaluated the gains of adopting NOMA schemes in terms of network throughput [20].

The rest of the paper is organized as follows: Section II provides an overview of the relevant state-of-art regarding semi-GF NOMA schemes and describes this work's contributions. Furthermore, Section III introduces the system model and presents the priority-based SGF-NOMA protocol. The theoretical framework for the proposed protocol is presented in Section IV, while the analytical evaluation results are presented and discussed in Section V. Finally, a summary of the main findings and concluding remarks are provided in Section VI.

Notations: The absolute value, exponential and base-2 logarithm functions are respectively denoted by $|\cdot|$, $\exp(\cdot)$, and $\log_2(\cdot)$. Moreover, $\Pr(\mathcal{A})$ denotes the probability for event $\mathcal{A}$ to be valid, while $\mathcal{X} = \{x_1, x_2, \dots, x_L\}$ defines the set $\mathcal{X}$ containing L elements (i.e., $x_1, x_2, \dots, x_L$). Also, the functions $\min(\mathcal{X})$ and $\max(\mathcal{X})$ respectively return the minimum and maximum value of $\mathcal{X}$. Finally, $f_Y(\cdot)$ and $F_Y(\cdot)$ stand for the probability density function (PDF) and the cumulative distribution function (CDF) of random variable (RV) Y , respectively. Additionally, Table I summarizes the acronyms used throughout the paper.

II. RELATED WORKS AND CONTRIBUTIONS

From the performance analysis point of view, in [21], the outage probability (OP) of NOMA under partial channel information was extracted, while, in [22], the authors presented and analyzed the outage performance as well as the diversity order of cooperative NOMA. In [23], quantified the impact of phase noise on multi-carrier wireless systems that employ NOMA, by presenting closed-form expression for the corresponding OP, whereas, in [24], the authors evaluated the outage performance of NOMA-enabled systems in the presence of in-phase and quadrature imbalance. In addition, in [25], stochastic geometry tools were employed to derive the OP of NOMA-based downlink cloud radio access network, in which the remote radio heads were uniformly distributed and serve two paired users simultaneously. Meanwhile, a feasibility study for the application of NOMA in satellite communication networks was conducted in [26]. Similarly, in [27], the authors

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A part of this paper has been presented in [1].

TABLE I
ACRONYMS

Acronym	Definition	Acronym	Definition
AWGN	Additive white Gaussian noise	OP	Outage probability
BS	Base-station	PD	Primary device
CSI	Channel state information	PDF	Probability density function
CS	Compressive sensing	RRB	Radio resource block
CDF	Cumulative distribution function	RV	Random variable
DR	Data rate	SD	Secondary device
GB	Grant-based	SGF	Semi-grant-free
GF	Grant-free	SINR	Signal-to-interference-plus-noise-ratio
MUD	Multi-user detection	SNR	Signal-to-noise-ratio
NG-IoT	Next-generation internet-of-things	SAW	Stop-and-wait
NOMA	Non-orthogonal multiple access	SIC	Successive interference cancellation

characterized the sum capacity of NOMA-enabled underwater visible light communication networks.

The main drawback of NOMA is that, in the uplink, the base-station (BS) demands knowledge of both the set of active devices as well as their channel conditions. In several realistic scenarios of NG-IoT applications, this is not always feasible. As a response to this drawback, GF-NOMA was presented. The main idea behind GF-NOMA is to surpass the ALOHA approach in terms of reliability while achieving the same latency performance, by utilizing successive interference cancellation (SIC) and allowing multiple devices to use the same radio resource block (RRB) in a random fashion. The performance of GF-NOMA schemes has been the focus of several research works [28]–[29]. The authors of [28] proposed a dynamic compressive sensing (CS) multi-user-detection (MUD) scheme that exploits the temporal correlation of active devices and carried out an analysis in terms of performance. The respective performance analysis results show that the proposed scheme can achieve a higher performance in terms of bit-error-rate. In [30], the authors analyzed the performance of two uplink GF approaches over shared resources, namely a blind retransmission approach and a stop-and-wait (SAW) approach. The analysis reveals that the blind retransmission approach outperforms the SAW one in transmissions with low and medium frequencies, while the SAW approach is more efficient in high-frequency transmissions. Additionally, the authors of [15] developed a GF-NOMA framework that treats collisions as interference to the remaining received signals and derived simplified expressions for approximating the OP and network throughput. The analysis results show that SIC has a higher performance in terms of throughput compared to successive joint decoding and a similar performance in terms of OP.

In [31], Gharbieh *et al.* leveraged queuing theory and stochastic geometry to design a spatio-temporal analytical model for opportunistic uplink transmissions for wireless-powered IoT environments. The model can jointly capture the characteristics of temporal traffic generation, transmission success probability, and energy harvesting. Similarly, the authors of [32] used stochastic geometry to model and analyze a GF NOMA system that employs CS for the MUD process. Based on the proposed model, they derived closed-form expressions for evaluating the user detection probability, channel

estimation error, and total system DR. In [29], the authors proposed a priority-based GF access scheme that performs dynamic slot allocation to devices with different priority levels in a subframe-by-subframe manner. The proposed scheme can guarantee access to high-priority devices, while also offering a satisfactory performance to low-priority ones. In [33], the authors developed a tractable approach to derive the access delay violation probability in GF systems that utilize hybrid automatic repeat request schemes. Finally, the authors of [34] developed a beacon-aided slotted GF scheme for enabling massive device access and developed two algorithms that exploit the characteristic of sporadic uplink traffic to improve the device activity detection process.

Alternatively, semi-GF (SGF) access schemes have emerged to address the issues of GF access schemes, while also providing low-latency access to shared resources. The performance analysis of semi-GF-NOMA schemes has been the focus of several recent research works [35]–[41]. In more detail, in [35], the authors proposed types of SGF communication schemes, in which a device uses grant-based (GB) access to a channel, while the others are admitted through contention control mechanisms. Also, Yang *et al.* [36] proposed an adaptive power allocation solution for uplink SGF-NOMA networks that guarantees the requirements of GB devices. Moreover, the authors derived closed-form expressions and asymptotic analytical results of the OP of both grant-based and GF devices. The authors of [37] developed a SGF-NOMA uplink approach for multiple grant-based and GF devices. Also, a power control mechanism was integrated in order to improve the performance of GF devices. Furthermore, in [38], the authors proposed a new SGF scheme that opportunistically combines the SGF schemes presented in [35]. The scheme in [35] allows higher DRs and improved transmission robustness. In [39] the authors presented a performance analysis for a MIMO-assisted SGF scheme and derived closed-form expressions for the preamble detection accuracy and access success probability. Zhang *et al.* in [40] investigated the ergodic rates of a SGF NOMA system by deriving closed-form expressions for the ergodic rates of both grant-based and GF devices. In [41], stochastic geometry approaches are used to investigate the performance of SGF-NOMA networks, while a dynamic protocol is proposed that effectively reduces the GF interference experienced by GB devices. Additionally,

Lu *et al.* in [42] evaluated the outage performance of two SGF schemes, where the resource allocation is performance-oriented and fairness-oriented, respectively. In the considered schemes, one GB device and one GF device are scheduled over the same radio resources. The authors of [43] conducted an outage performance analysis for a SGF system that employs rate-splitting multiple access. The radio resources are shared between a GB and a GF device. Finally, Bai and Gu [44] proposed a mathematical framework that leverages stochastic geometry to analyze the association success probability when two GF devices share radio resources with a single GB device.

Despite the paramount importance of GF and SGF NOMA, non of the aforementioned approaches enables the co-existence of scheduled and random access devices of diverse data-rate requirements. Motivated by this, the current contribution is devoted to the presentation of a novel SGF-NOMA, that increases the network capacity while ensuring reliable connectivity of two both scheduled and random access devices. Particularly, in the proposed protocol, two device categories exist, namely primary and secondary devices (SDs). Due to their increased DR requirements, primary devices (PDs) require scheduled or highly-probable random access, whereas SDs have very low DR requirements and access the network in a random manner. Moreover, the BS creates RRBs based on the number of PDs, while primary and two SDs are scheduled in the same RRB. The scheduling of devices to a RRB is determined by their respective DR requirements and by the transmission probability of the SDs. To validate the feasibility and efficiency of the proposed protocol, we carry out a performance analysis to derive closed-form expressions for the respective OPs and throughput of all devices.

III. SYSTEM MODEL AND PROTOCOL DESCRIPTION

A. System Model

As illustrated in Fig. 1, we consider an uplink scenario, where a single BS provides access to N devices. We assume that $M \leq N$ PDs request high-DR continuous connectivity, while, the rest $N - M$ SDs request low-DR event-driven connectivity. This scenario is considered realistic and common in NG-IoT environments, where video streaming, two and/or three-dimensional extended reality, and other high-DR applications, need to coexist with notification and alert-providing sensors, which require considerably lower DR [45], [46].

In what follows, we use

$$\mathcal{P} = \{P_1, P_2, \dots, P_M\} \quad (1)$$

to define the set of PDs. Note that in (1), P_m with $m \in [1, M]$ stands for the index of the m -th PD. Similarly,

$$\mathcal{S} = \{S_1, S_2, \dots, S_{N-M}\} \quad (2)$$

defines the set of SDs. In (2), S_n with $n \in [1, N - M]$ denotes the index of the n -th SD. Let us also employ

$$\mathcal{R}_P = \{R_{P_1}, R_{P_2}, \dots, R_{P_M}\} \quad (3)$$

and

$$\mathcal{R}_S = \{R_{S_1}, R_{S_2}, \dots, R_{S_{N-M}}\} \quad (4)$$

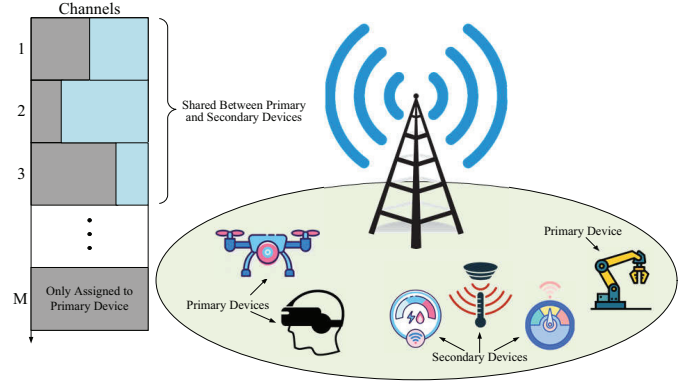


Fig. 1. Uplink priority-based SGF NOMA Scenario

to define the sets of the DR requirements in bits/s/Hz for the PDs and SDs, respectively. In more detail, R_{P_m} and R_{S_n} with $m \in [1, M]$ and $n \in [1, N - M]$ respectively stand for the DR requirement of P_m and S_n . Notice, that in real-world deployments, the following inequality is expected to hold:

$$\min(\mathcal{R}_P) > \max(\mathcal{R}_S). \quad (5)$$

Finally, we use

$$\mathcal{P}_S = \{P_{S_1}, P_{S_2}, \dots, P_{S_{N-M}}\} \quad (6)$$

to define the set of SDs transmission probability. The BS creates the set of

$$\mathcal{R} = \{R_1, R_2, \dots, R_M\} \quad (7)$$

of M RRBs and in each R_m , with $m \in [1, M]$, allocates a PD as well as a predetermined number of SDs, in a way that the DR requirements of all the allocated devices are ideally satisfied. For simplicity, we set the number of the maximum SDs that can be allocated in a single RRB to two. It is assumed that P_m and S_k and S_l with $m \in [1, M]$ and $k \neq l \in [1, N - M]$ are allocated to R_m . Thus, in the information transmission phase, the received signal from the R_m at the BS can be expressed as

$$r_m = h_{P_m} s_{P_m} + \theta_k h_{S_k} s_{S_k} + \theta_l h_{S_l} s_{S_l} + n_m, \quad (8)$$

where h_{P_m} , h_{S_k} , and h_{S_l} are the channel coefficients of the P_m -BS, S_k -BS, and S_l -BS links, respectively. Moreover, s_{P_m} , s_{S_k} , and s_{S_l} are the transmission signals by the P_m , S_k , and S_l . Likewise, n_m denotes the additive white Gaussian noise (AWGN) at R_m . Finally,

$$\theta_{\{i\}} = \begin{cases} 1, & S_i \text{ transmits} \\ 0, & \text{otherwise} \end{cases}, \quad (9)$$

with $i \in \{k, l\}$ and $k \neq l \in [1, N - M]$.

The BS decodes s_{P_m} first by treating the received signals by S_k and S_l as interference. Thus, the received signal-to-interference-plus-noise-ratio (SINR) for P_m can be obtained as

$$\gamma_{P_m} = \frac{|h_{P_m}|^2 E_{P_m}}{\theta_k |h_{S_k}|^2 E_{S_k} + \theta_l |h_{S_l}|^2 E_{S_l} + N_o}, \quad (10)$$

where E_{P_m} , E_{S_k} , and E_{S_l} are the transmission powers of P_m , S_k , and S_l , respectively, while N_o stands for the noise power.

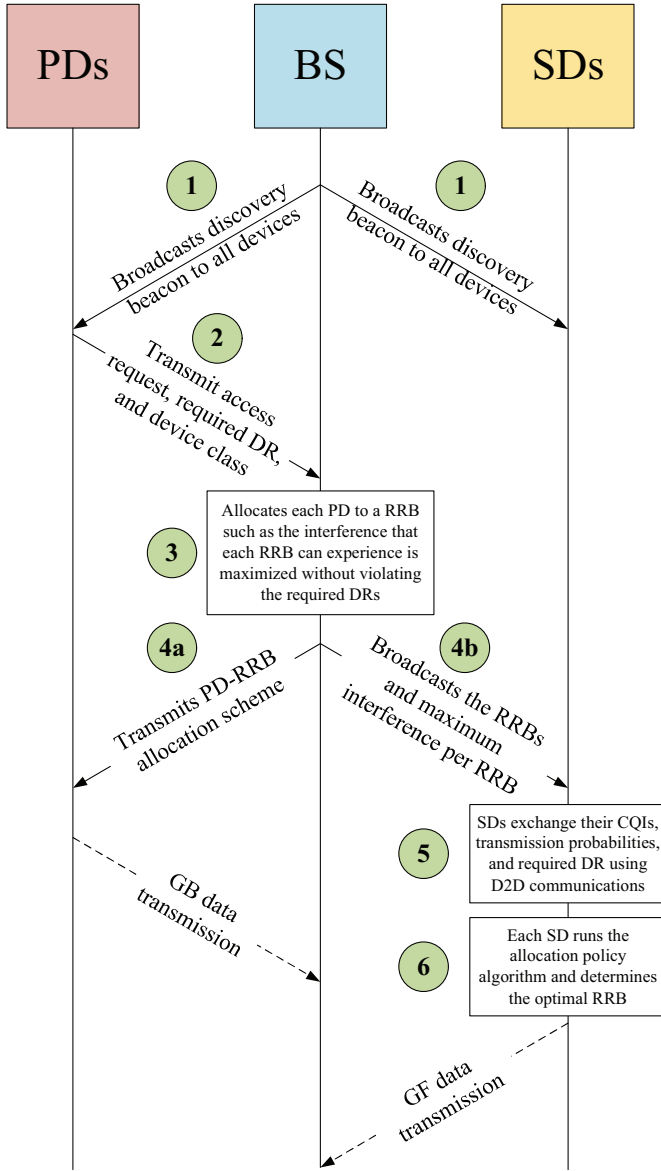


Fig. 2. Priority-based SGF-NOMA protocol

It is assumed that $|h_{S_k}| > |h_{S_l}|$; hence, according to the uplink NOMA principle [47], $E_{S_k} > E_{S_l}$. As a consequence, the BS will next decode the received by the S_k device message. The SINR for S_k can be expressed as

$$\gamma_{S_k} = \frac{\theta_k |h_{S_k}|^2 E_{S_k}}{\theta_l |h_{S_l}|^2 E_{S_l} + N_o}. \quad (11)$$

Finally, the SNR for S_l can be written as

$$\gamma_{S_l} = \frac{\theta_l |h_{S_l}|^2 E_{S_l}}{N_o}. \quad (12)$$

B. Protocol Description

The protocol flow is depicted in Fig. 2, along with a mapping of the respective steps. The protocol consists of three phases, namely the discovery phase, the PDs GB access phase, and the SDs GF access phase. During the discovery

phase, the BS broadcasts a synchronization beacon to notify all devices of its presence (Step 1). The beacon includes BS identification information and pilot symbols for channel estimation. Upon receiving the beacon, all devices carry out the channel estimation process. Then, each PD transmits an access request signal, along with its DR requirements (Step 2). Using that signal, the BS gains knowledge of the number of PDs, their DR requirements, and their respective channel state information (CSI).

In the PDs GB access phase, the BS generates a number of RRBs that is equal to the number of PDs (i.e., M) and allocates a single PD to each RRB in a way that the achievable DR is maximized and is greater than the required DR (Step 3). Next, the BS transmits an access grant signal to the PDs containing the PD-RRB allocation information (Step 4a). In addition, the BS broadcasts the available RRBs and the maximum interference that can be introduced to each RRB without violating the corresponding PD's required DR (Step 4b).

During the SDs GF access phase, the SDs receive the information about the available RRBs and broadcast their DR requirements their CSI, and their transmission probabilities through a secondary cognitive radio network that is formed among the SDs (Step 5). As a result, all SDs are aware of the DR requirements, the CSI, and transmission probabilities of the rest SDs, as well as the level of interference that each RRB can endure. Using this knowledge, each SD independently executes the same allocation policy algorithm in order to determine the RRB that can be used to transmit their data (Step 6). Specifically, the allocation policy considers the transmission probability of SDs as well as the DR requirements of both PDs and SDs, giving priority to the PDs' DR requirements. Different resource allocation algorithms can be utilized, based on the aim of the allocation policy (e.g., minimizing the total energy consumption, maximizing spectral and/or energy efficiency, maximizing system fairness, maximizing the total system throughput, etc.) [48], [49].

Of note, the BS performs successive interference cancellation (SIC) to distinguish the device signals. Also, since alternative channels are used for the SDs communications, in the SDs GF access phase, the signaling overhead between the BS and the SDs is minimized. In addition, the proposed protocol assumes that each device has been pre-configured as PD or SD. Consequently, based on its type, the device will be involved in different phases and steps. Specifically, PDs will be involved in Steps 2, 3, and 4a, while SDs will be involved in Steps 4b, 5, and 6. Finally, it is assumed that the DR requirements, SNR thresholds, and transmission probabilities remain steady during a particular communication cycle.

IV. PERFORMANCE ANALYSIS

A. Outage Probability Analysis

In this section, the theoretical framework for the evaluation of the system OP is provided. Theorems 1–4 respectively return the OPs for P_m , S_k and S_l for the following cases: i) Case 1: Only P_m uses R_n , ii) Case 2: Both P_m and S_k use R_n , iii) Case 3: Both P_m and S_l use R_n , and iv) Case 4: P_m , S_k , and S_l use R_n .

Theorem 1. In the case in which R_n is used only by the primary user P_m , the OP of P_m can be evaluated as

$$P_{o,P_m}^{C_1}(\gamma_{th}^{P_m}) = 1 - \exp\left(-\frac{\gamma_{th}^{P_m} N_o}{2E_{P_m}}\right), \quad (13)$$

where $\gamma_{th}^{P_m}$ is the SNR threshold of P_m .

Proof: For brevity, the proof of the Theorem is provided in Appendix A. ■

Theorem 2. In the case in which R_n is used by both P_m and S_k , the OP of P_m can be obtained as

$$P_{o,P_m}^{C_2}(\gamma_{th}^{P_m}) = 1 - \frac{2}{2 + \frac{\gamma_{th}^{P_m} E_{S_k}}{E_{P_m}}} \exp\left(-\frac{\gamma_{th}^{P_m} N_o}{2E_{P_m}}\right) \quad (14)$$

while, the OP of S_k can be expressed as

$$\begin{aligned} P_{o,S_k}^{C_2}(\gamma_{th}^{S_k}) &= 2 - \exp\left(-\frac{1}{2} \left(\frac{N_o \gamma_{th}^{P_m}}{E_{P_m}}\right)^2\right) \\ &- \exp\left(-\frac{\gamma_{th}^{S_k} N_o}{E_{S_k}}\right) + \frac{2\gamma_{th}^{P_m} E_{S_k}}{2E_{P_m} + \gamma_{th}^{P_m} E_{S_k}} \\ &\times \exp\left(-\frac{N_o^2 \gamma_{th}^{P_m} (2E_{P_m} + \gamma_{th}^{P_m} E_{S_k}) - 4N_o}{E_{P_m}^2} \right) \end{aligned} \quad (15)$$

where $\gamma_{th}^{S_k}$ stands for the SNR threshold of S_k .

Proof: For brevity, the proof of the Theorem is provided in Appendix B. ■

Theorem 3. In the case in which R_n is used by both P_m and S_l , the OP of P_m can be obtained as

$$P_{o,P_m}^{C_3}(\gamma_{th}^{P_m}) = 1 - \frac{E_{P_m}^2 \exp\left(-\gamma_{th}^{P_m} \frac{N_o}{E_{P_m}}\right)}{\left(E_{P_m} + E_{S_l} \gamma_{th}^{P_m}\right) \left(E_{P_m} + 2E_{S_l} \gamma_{th}^{P_m}\right)} \quad (16)$$

while the OP of S_l can be obtained as

$$\begin{aligned} P_{o,S_l}^{C_3}(\gamma_{th}^{S_l}) &= 1 - \exp\left(-\left(\frac{N_o \gamma_{th}^{P_m}}{\sqrt{2}E_{P_m}}\right)^2\right) \\ &+ \left(\exp\left(\frac{N_o \gamma_{th}^{S_l}}{2E_{S_l}}\right) - 1\right)^2 \\ &+ \frac{4\gamma_{th}^{P_m} E_{S_l} \exp\left(-\frac{N_o^2 \gamma_{th}^{P_m} (E_{P_m} + \gamma_{th}^{P_m} E_{S_l}) - 2N_o}{E_{P_m}^2} \right)}{E_{P_m} + \gamma_{th}^{P_m} E_{S_l}} \\ &- \frac{2\gamma_{th}^{P_m} E_{S_l} \exp\left(-\frac{4N_o - \frac{N_o^2 \gamma_{th}^{P_m} (E_{P_m} + \gamma_{th}^{P_m} E_{S_l})}{E_{P_m}^2}}{4E_{S_l}} \right)}{2E_{P_m} + \gamma_{th}^{P_m} E_{S_l}} \end{aligned} \quad (17)$$

where $\gamma_{th}^{S_l}$ denotes the SNR threshold of S_l .

Proof: For brevity, the proof of the Theorem is provided in Appendix C. ■

Theorem 4. In the case in which R_n is used by P_m , S_k , and S_l , the OP of P_m , S_k and S_l can be respectively obtained as

$$P_{o,P_m}^{C_4}(\gamma_{th}^{P_m}) = 1 - \frac{4E_{P_m}^3}{Q_{P_m}^{C_4}} \exp\left(-\frac{\gamma_{th}^{P_m} N_o}{2E_{P_m}}\right), \quad (18)$$

where $Q_{P_m}^{C_4}$ can be expressed as

$$\begin{aligned} Q_{P_m}^{C_4} &= \left(2E_{P_m} + \gamma_{th}^{P_m} E_{S_k}\right) \left(E_{P_m} + \gamma_{th}^{P_m} E_{S_l}\right) \\ &\times \left(2E_{P_m} + \gamma_{th}^{P_m} E_{S_l}\right) \end{aligned} \quad (19)$$

Moreover,

$$\begin{aligned} P_{o,S_k}^{C_4}(\gamma_{th}^{S_k}) &= 1 - 2 \exp\left(-\frac{N_o \gamma_{th}^{P_m}}{2E_{P_m}}\right) Q_{S_k}^{C_4} + \exp\left(-\frac{\gamma_{th}^{S_k}}{E_{S_k}}\right) \\ &\times E_{S_k} \left(\frac{4 \exp\left(\frac{\gamma_{th}^{S_k}}{2E_{S_k}}\right)}{2E_{S_k} + \gamma_{th}^{S_k} E_{S_l}} + \frac{1}{E_{S_k} + \gamma_{th}^{S_k} E_{S_l}}\right) \end{aligned} \quad (20)$$

where $Q_{S_k}^{C_4}$ can be written as

$$\begin{aligned} Q_{S_k}^{C_4} &= \frac{\gamma_{th}^{P_m} E_{P_m}^2 E_{S_k}}{\left(E_{P_m} + \gamma_{th}^{P_m} E_{S_k}\right) \left(E_{P_m} + \gamma_{th}^{P_m} E_{S_l}\right)} \\ &\times \frac{1}{\left(2E_{P_m} + \gamma_{th}^{P_m} E_{S_l}\right)} \end{aligned} \quad (21)$$

Finally,

$$\begin{aligned} P_{o,S_l}^{C_4}(\gamma_{th}^{S_l}) &= \frac{E_{S_l}^2 \exp\left(\frac{N_o}{E_{S_l}}\right)}{\left(E_{S_k} + E_{S_l}\right) \left(2E_{S_k} + E_{S_l}\right)} \\ &+ \left(\frac{N_o \gamma_{th}^{S_l} \exp\left(-\frac{N_o \gamma_{th}^{S_l}}{E_{S_l}}\right)}{E_{S_l}} - 1\right)^2 \end{aligned} \quad (22)$$

Proof: For brevity, the proof of the Theorem is provided in Appendix D. ■

Proposition 1. The respective OPs of a PD (i.e., P_m) and two SDs (i.e., S_k and S_l) sharing the m -th channel are derived as

$$\begin{aligned} P_{o,P_m}(\gamma_{th}^{P_m}) &= P_{o,P_m}^{C_1}(\gamma_{th}^{P_m}) (1 - P_{S_k})(1 - P_{S_l}) \\ &+ P_{o,P_m}^{C_2}(\gamma_{th}^{P_m}) P_{S_k}(1 - P_{S_l}) \\ &+ P_{o,P_m}^{C_3}(\gamma_{th}^{P_m}) (1 - P_{S_k})P_{S_l} \\ &+ P_{o,P_m}^{C_4}(\gamma_{th}^{P_m}) P_{S_k}P_{S_l} \end{aligned} \quad (23)$$

$$\begin{aligned} P_{o,S_k}(\gamma_{th}^{S_k}) &= P_{o,S_k}^{C_2}(\gamma_{th}^{S_k}) P_{S_k}(1 - P_{S_l}) \\ &+ P_{o,S_k}^{C_4}(\gamma_{th}^{S_k}) P_{S_k}P_{S_l} \end{aligned} \quad (24)$$

$$P_{o,S_l}(\gamma_{th}^{S_l}) = P_{o,S_l}^{C_3}(\gamma_{th}^{S_l}) (1 - P_{S_k}) P_{S_l} + P_{o,S_l}^{C_4}(\gamma_{th}^{S_l}) P_{S_k} P_{S_l} \quad (25)$$

Proof: Let $\mathcal{C}_j$ with $j \in [1, 4]$ denote the event that case j is valid. Then, the OP of P_m , S_k and S_l can be respectively obtained as

$$P_{o,P_m}(\gamma_{th}^{P_m}) = \Pr(\mathcal{C}_1) P_{o,P_m}^{C_1}(\gamma_{th}^{P_m}) + \Pr(\mathcal{C}_2) P_{o,P_m}^{C_2}(\gamma_{th}^{P_m}) + \Pr(\mathcal{C}_3) P_{o,P_m}^{C_3}(\gamma_{th}^{P_m}) + \Pr(\mathcal{C}_4) P_{o,P_m}^{C_4}(\gamma_{th}^{P_m}), \quad (26)$$

$$P_{o,S_k}(\gamma_{th}^{S_k}) = \Pr(\mathcal{C}_2) P_{o,S_k}^{C_2}(\gamma_{th}^{S_k}) + \Pr(\mathcal{C}_4) P_{o,S_k}^{C_4}(\gamma_{th}^{S_k}) \quad (27)$$

and

$$P_{o,S_l}(\gamma_{th}^{S_l}) = \Pr(\mathcal{C}_3) P_{o,S_l}^{C_3}(\gamma_{th}^{S_l}) + \Pr(\mathcal{C}_4) P_{o,S_l}^{C_4}(\gamma_{th}^{S_l}), \quad (28)$$

where

$$\Pr(\mathcal{C}_1) = (1 - P_{S_k})(1 - P_{S_l}), \quad (29)$$

$$\Pr(\mathcal{C}_2) = P_{S_k}(1 - P_{S_l}), \quad (30)$$

$$\Pr(\mathcal{C}_3) = (1 - P_{S_k})P_{S_l} \quad (31)$$

and

$$\Pr(\mathcal{C}_4) = P_{S_k}P_{S_l}. \quad (32)$$

By substituting (29)–(32) to (26)–(28), we obtain (23) – (25). This concludes the proof. ■

B. Throughput Analysis

In this subsection, an analysis of each device throughput is presented. For convenience, we assume that the maximum number of SDs that can be connected at each RRB are 2. In this case, the throughput of P_m can be evaluated as

$$T_{P_m} = \log_2 \left(1 - \gamma_{th}^{P_m} \right) \left(1 - F_{\gamma_{P_m}} \left(\gamma_{th}^{P_m} \right) \right) \quad (33)$$

where $\gamma_{th}^{P_m}$ is the PD's signal-to-noise-ratio (SNR) threshold, while $F_{\gamma_{P_m}} \left(\gamma_{th}^{P_m} \right)$ is the CDF of the PD's instantaneous SNR and can be obtained as

$$F_{\gamma_{P_m}} \left(\gamma_{th}^{P_m} \right) = P_{o,P_m}^{C_1} \left(\gamma_{th}^{P_m} \right) (1 - P_{S_k})(1 - P_{S_l}) + P_{o,P_m}^{C_2} \left(\gamma_{th}^{P_m} \right) P_{S_k}(1 - P_{S_l}) + P_{o,P_m}^{C_3} \left(\gamma_{th}^{P_m} \right) (1 - P_{S_k})P_{S_l} + P_{o,P_m}^{C_4} \left(\gamma_{th}^{P_m} \right) P_{S_k}P_{S_l}, \quad (34)$$

Similarly, the throughput of S_k can be obtained as

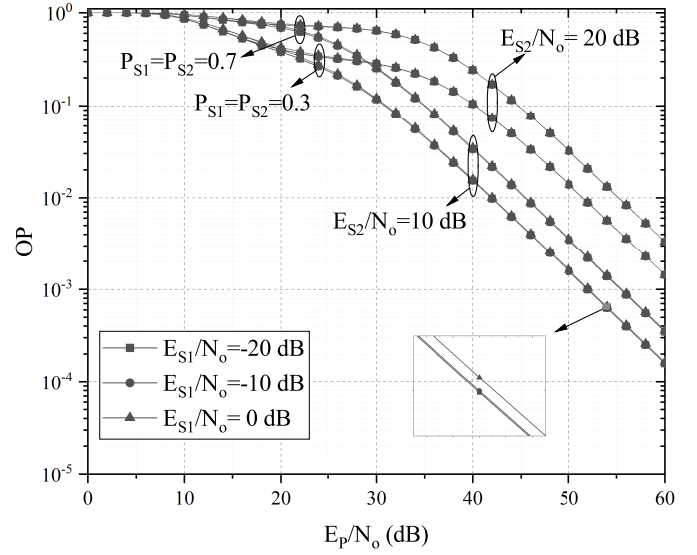


Fig. 3. OP of the PD vs E_P/N_o for different values of E_{S1} , E_{S2} , P_{S1} , and P_{S2}

$$T_{S_k} = \log_2 \left(1 - \gamma_{th}^{S_k} \right) \left(1 - F_{\gamma_{S_k}} \left(\gamma_{th}^{S_k} \right) \right) \quad (35)$$

where $F_{\gamma_{S_k}} \left(\gamma_{th}^{S_k} \right)$ is the CDF of the S_k 's instantaneous SNR that can be obtained by

$$P_{o,S_k} \left(\gamma_{th}^{S_k} \right) = P_{o,S_k}^{C_2} \left(\gamma_{th}^{S_k} \right) P_{S_k}(1 - P_{S_l}) + P_{o,S_k}^{C_4} \left(\gamma_{th}^{S_k} \right) P_{S_k}P_{S_l} \quad (36)$$

Finally, the throughput of S_l can be expressed as

$$T_{S_l} = \log_2 \left(1 - \gamma_{th}^{S_l} \right) \left(1 - F_{\gamma_{S_l}} \left(\gamma_{th}^{S_l} \right) \right) \quad (37)$$

where $F_{\gamma_{S_l}} \left(\gamma_{th}^{S_l} \right)$ is the CDF of the S_l 's instantaneous SNR which is obtained by

$$P_{o,S_l} \left(\gamma_{th}^{S_l} \right) = P_{o,S_l}^{C_3} \left(\gamma_{th}^{S_l} \right) (1 - P_{S_k})P_{S_l} + P_{o,S_l}^{C_4} \left(\gamma_{th}^{S_l} \right) P_{S_k}P_{S_l}. \quad (38)$$

V. RESULTS & DISCUSSION

This section is focused on presenting the numerical results accompanied by insightful discussions that quantify the outage and throughput performance of the priority-based SGF-NOMA protocol and verify the theoretical framework as well as provide useful guidelines for the protocol deployment. The following scenario is considered: we assume that a single PD and two SDs have been assigned to the same RRB. Unless otherwise stated, we set $\gamma_{th}^{P_m}$ to be equal to 1.5 dB and $\gamma_{th}^{S_k} = \gamma_{th}^{S_l} = 0$ dB¹. For simplicity in what follows, we omit m and we set $k = 1$ and $l = 2$.

Figure 3 depicts the OP of the PD as a function of E_P/N_o for different values of E_{S1} , E_{S2} , P_{S1} , and P_{S2} . As expected,

¹A SNR threshold of 0dB can lead to higher than 1 bps/Hz, which can efficiently satisfy the requirements of low-DR IoT applications.

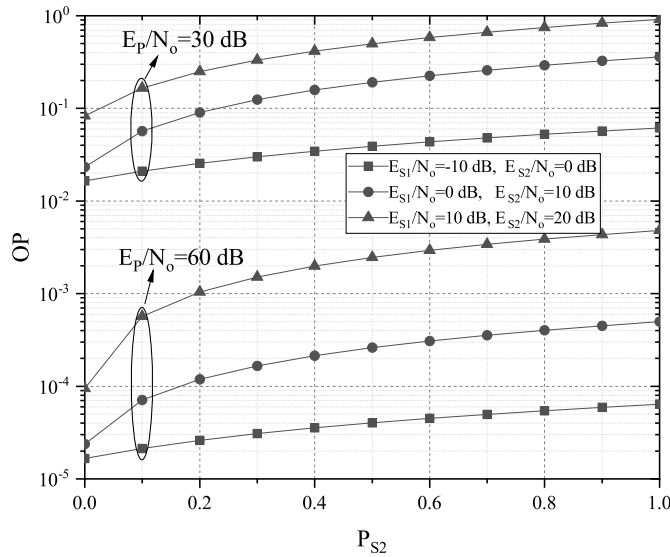


Fig. 4. OP of the PD vs P_{S_2} for different values of E_P , E_{S_1} , and E_{S_2}

for fixed E_{S_1} , E_{S_2} , P_{S_1} and P_{S_2} , as E_P/N_o increases, the OP of the PD decreases. For instance, for $E_{S_1}/N_o = -20$ dB and $P_{S_1} = P_{S_2} = 0.3$, the OP of the PD improves by approximately one order of magnitude, as E_P/N_o increases from 30 to 40 dB. Additionally, for given E_P/N_o , E_{S_1}/N_o , P_{S_1} and P_{S_2} , as E_{S_2}/N_o increases, the interference to the PD by S_2 increases; thus, the OP of the PD also increases. For instance, for $E_P/N_o = 40$ dB, $E_{S_1}/N_o = 0$ dB, $P_{S_1} = P_{S_2} = 0.3$, the OP increases for about 10 times, as E_{S_2}/N_o increases from 10 to 20 dB. This reveals the detrimental effect of S_2 interference level at the outage performance of PD. Similarly, for given E_P/N_o , E_{S_2}/N_o , P_{S_1} and P_{S_2} , as E_{S_1}/N_o increases, the interference to the PD by S_1 increases; as a consequence, an outage performance degradation is observed. For example, for $E_P/N_o = 40$ dB, $E_{S_2}/N_o = 10$ dB, $P_{S_1} = P_{S_2} = 0.3$, the OP increases for about 5 times, as E_{S_1}/N_o increases from -10 to 0 dB. From this example, it becomes evident that the same transmission SNR increase at both S_1 and S_2 results in different outage performance degradation. This is expected due to the fact that $|h_{S_1}| > |h_{S_2}|$. Moreover, from this figure, it becomes apparent that, for fixed E_P/N_o , E_{S_1}/N_o , and E_{S_2}/N_o , as $P_{S_1} = P_{S_2}$ increases, the probability of interference to PD by S_1 and S_2 increases; hence, the OP for PD increases. For instance, $E_P/N_o = 30$ dB, $E_{S_1}/N_o = -20$ dB, and $E_{S_2}/N_o = 10$ dB, the OP of the PD increases by approximately 2.5 times, as $P_{S_1} = P_{S_2}$ increases from 0.2 to 0.7. This highlights the importance of accurately capturing the transmission probabilities of SDs, when analyzing and deploying the priority-based SGF NOMA protocol.

Figure 4 illustrates the OP of the PD as a function of P_{S_2} for different values of E_P/N_o , E_{S_1}/N_o , and E_{S_2}/N_o , assuming that $P_{S_1} = 0.5$. Notice that in the special case in which $P_{S_2} = 0$, no second SD uses the RRB. On the other hand, for $P_{S_2} = 1$, S_2 continuously transmits to the RRB. These two special cases respectively correspond to the best and worst scenarios for the OP of the PD. From this figure,

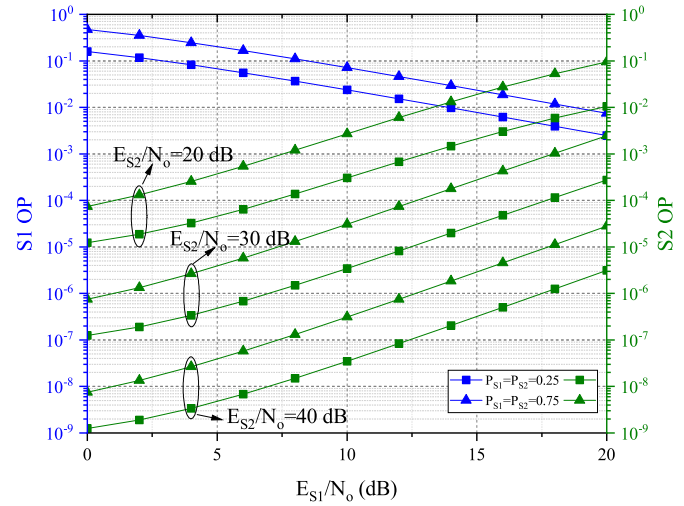


Fig. 5. OP of SDs vs E_{S_1}/N_o for different values of E_{S_2} , P_{S_1} and P_{S_2}

we observe that, for given E_P/N_o , E_{S_1}/N_o , and E_{S_2}/N_o , as P_{S_2} increases, the probability of S_2 to cause interference to the PD increases; as a result, the OP also increases. Moreover, for fixed P_{S_2} , E_{S_1}/N_o , and E_{S_2}/N_o , as E_P/N_o increases, the received signal strength of the PD desired signal increases; thus, the average received SINR increases, which results to an outage performance enhancement. For example, for $P_{S_2} = 0.2$, $E_{S_1} = 10$ dBm and $E_{S_2} = 20$ dBm, the OP of the PD changes from $10^{-0.5}$ to 10^{-2} as E_P increases from 30 to 60 dBm. Likewise, for fixed P_{S_2} , E_P/N_o , and E_{S_1}/N_o , as E_{S_2}/N_o increases, the interference caused by S_2 increases; as a consequence, the outage performance degrades. Finally, for given P_{S_2} , E_P/N_o , and E_{S_2}/N_o , as E_{S_1}/N_o increases, the interference caused by S_1 increases; as a consequence, the OP of the PD also increases.

Figure 5 presents the OPs of the SDs as functions of E_{S_1}/N_o for different values of E_{S_2}/N_o , P_{S_1} , and P_{S_2} , assuming $E_P/N_o =$. For given E_{S_2}/N_o , P_{S_1} and P_{S_2} , as E_{S_1}/N_o increases, the OP of S_1 decreases, while the OP of S_2 increases. This is expected since, as the transmission SNR of S_1 increases, the average received SINR at S_1 increases, while the average received SINR at S_2 decreases. Moreover, for given E_{S_1}/N_o , P_{S_1} and P_{S_2} , as E_{S_2} increases, the interference caused by S_2 to S_1 increases; thus the OP of S_1 increases. On the other hand, for given E_{S_1}/N_o , P_{S_1} and P_{S_2} , as E_{S_1}/N_o increases, the desired signal strength at S_2 increases; thus, the average SNR of s_2 at S_2 increases and in turn the OP of S_2 decreases. In addition, for fixed E_{S_1}/N_o , P_{S_1} and P_{S_2} , as E_{S_2}/N_o , the OP of S_2 decreases. Finally, for given E_{S_1}/N_o , and E_{S_2}/N_o , as $P_{S_1} = P_{S_2}$ increases, both the OP of S_1 and S_2 increase.

Figure 6 shows the PD throughput as a function of the power-to-noise gain for different SD transmission probabilities. The power-to-noise gain of the PD is increased from 0 to 50 dB, while the power-to-noise gains of S_1 and S_2 are set to 0 and 10 dB, respectively. Also, the SNR threshold of both SDs is set to -5 dB. The PD throughput is evaluated for $\gamma_{th}^P = 10$ dB and $\gamma_{th}^P = 20$ dB, and for three cases of SD

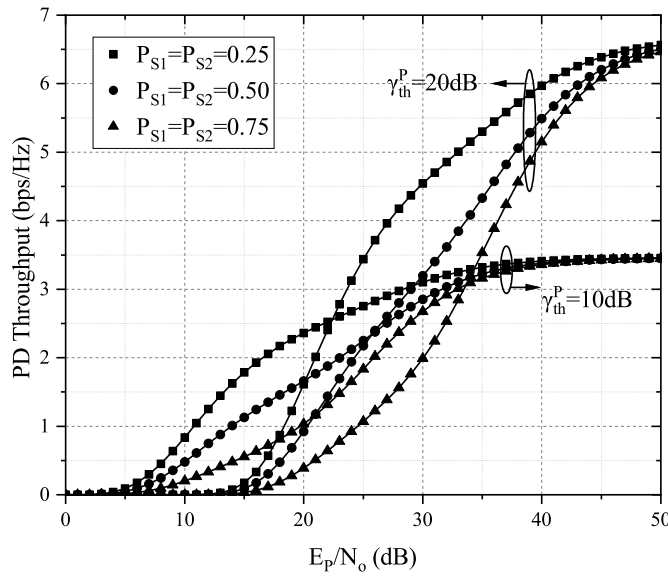


Fig. 6. PD throughput vs power-to-noise gain for various SDs transmission probabilities

transmission probabilities, namely 0.25, 0.50, and 0.75.

As expected, the PD throughput increases as the PD power-to-noise gain is increased. For example, for $\gamma_{th}^P = 10\text{dB}$, $P_{S1} = P_{S2} = 0.25$, and $E_P/N_o = 10\text{dB}$, the PD throughput is 1 bps/Hz. When the power-to-noise gain is increased to $E_P/N_o = 50\text{dB}$, the PD throughput is 3.5 bps/Hz. On the other hand, for the case of $\gamma_{th}^P = 20\text{dB}$, the respective PD throughput when $E_P/N_o = 10\text{dB}$ is about 0 bps/Hz. However, when the power-to-noise gain is increased to $E_P/N_o = 50\text{dB}$, the corresponding PD throughput is 6.5 bps/Hz. This indicates that higher γ_{th}^P values lead to lower throughput when the E_P/N_o ratio is low since the PD will experience more outages. Nevertheless, for higher power-to-noise gains, the PD will achieve much higher throughput.

Concerning the transmission probabilities of the SDs, when the P_{S1} and P_{S2} are increased, the respective PD throughput will decrease as the SDs will introduce higher interference. For example, for $\gamma_{th}^P = 20\text{dB}$ and $E_P/N_o = 30\text{dB}$, when $P_{S1} = P_{S2} = 0.25$, the PD throughput is about 4.5 bps/Hz, while when $P_{S1} = P_{S2} = 0.75$, the PD throughput is about 2 bps/Hz. In addition, for high values of E_P/N_o , the respective throughput of all cases converges, as the power-to-noise gain is high enough to minimize the experienced interference.

Figure 7 depicts the PD throughput as a function of the SNR threshold for various PD power-to-noise gains and SDs transmission probabilities. The SNR threshold of the PD is increased from 0 to 25 dB, while the PD power-to-noise gains are set to 10, 20, and 30 dB. Moreover, the SNR thresholds of SDs are set to -5 dB. Also, the PD throughput is evaluated for three cases of SDs transmission probabilities, namely 0.25, 0.50, and 0.75.

As γ_{th}^P is increased, the PD throughput increases up to a particular value and then it starts decreasing. This is expected since, for higher γ_{th}^P values, the device outage probability will have a higher impact on the throughput than the data

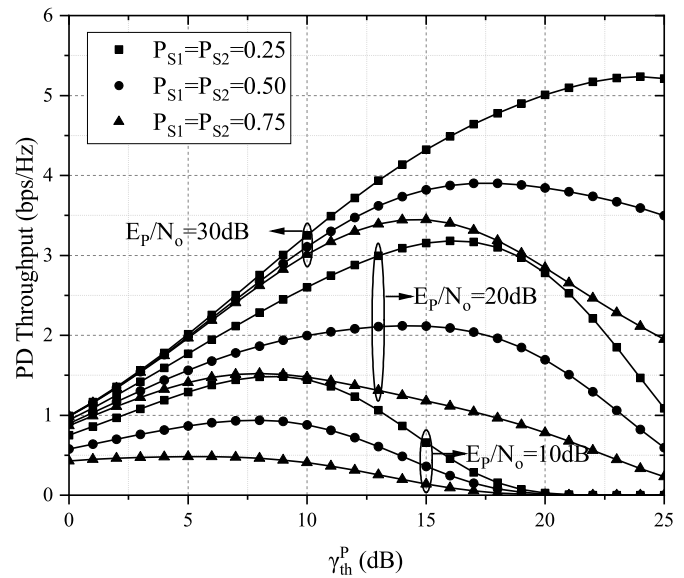


Fig. 7. PD throughput vs SNR threshold for various values of E_P/N_o and $P_{S1} = P_{S2}$

rate. Considering the PD power-to-noise gain, higher values of gains lead to higher overall throughput. Also, when $E_P/N_o = 10\text{dB}$, the maximum PD throughput is achieved at $\gamma_{th}^P = 8\text{dB}$, while the corresponding maximum PD throughput when $E_P/N_o = 30\text{dB}$ is achieved at $\gamma_{th}^P = 24\text{dB}$. This reveals that higher power-to-noise gains lead to decreased PD outage probability. Of note, for $E_P/N_o = 10\text{dB}$, when $\gamma_{th}^P > 20\text{dB}$, the PD throughput is almost zero since the power-to-noise gain is not enough to achieve the required SNR threshold.

Moreover, when the P_{S1} and P_{S2} are increased, the respective PD throughput will decrease because of the increased interference caused by the SDs. For example, for $E_P/N_o = 20\text{dB}$ and $\gamma_{th}^P = 10\text{dB}$, when $P_{S1} = P_{S2} = 0.25$, the PD throughput is about 1 bps/Hz, while when $P_{S1} = P_{S2} = 0.75$, the PD throughput is about 3 bps/Hz. Furthermore, for higher values of SNR threshold, the PD throughput in all cases is expected to converge to 0 bps/Hz as the outage probability of the device will continue to increase.

Finally, the respective throughput of S_1 and S_2 as functions of the SNR threshold for various transmission probabilities are illustrated in Fig. 8. The SNR threshold is increased from -5 to 15 dB, while the transmission probabilities of both SDs are set to 0.25, 0.50, and 0.75. Moreover, the power-to-noise gains of SD1 and SD2 are respectively set to 0 and 10 dB.

Similarly to the PD case, the SD throughput increases up to a particular value and then it starts decreasing, as the SNR threshold of the SDs is increased, due to the higher impact of the outage probability. As $E_{S1}/N_o = 0\text{dB}$, the throughput of S_1 for all cases of transmission probabilities will converge to 0 bps/Hz since the power-to-noise gain will not be able to achieve the required SNR threshold. On the other hand, since $E_{S2}/N_o = 10\text{dB}$, S_2 will achieve the required SNR threshold in the considered SNR range.

Finally, the respective S_1 and S_2 throughput decreases as the transmission probabilities of the SDs are increased.

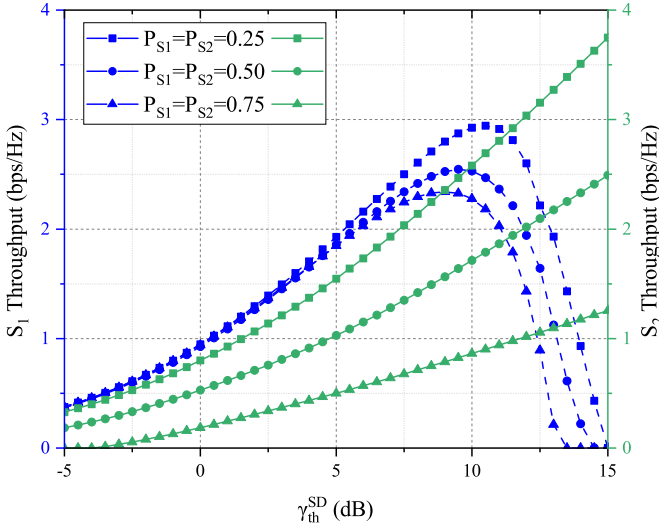


Fig. 8. S_1 and S_2 throughput vs SNR threshold for various $P_{S1} = P_{S2}$

This happens because of the increased interference caused by the higher SD transmission frequency. For example, when $\gamma_{th}^{SD} = 10$ dB, the S_1 throughput is 3, 2.5, and 2.2 bps/Hz for $P_{S1} = P_{S2} = 0.25$, $P_{S1} = P_{S2} = 0.50$, and $P_{S1} = P_{S2} = 0.75$, respectively. On the other hand, the corresponding S_2 throughput for the same cases is 2.5, 1.6, and 0.9 bps/Hz, respectively.

The aforementioned results are derived through a numerical analysis using the equations derived in Section IV. In this respect, the expected outage probabilities and device throughput can be determined by changing each of the functions' variables (e.g., required threshold, power budget, transmission probabilities). In real-world systems, the outage probabilities can be evaluated by configuring the BS to periodically record the received SNR and calculate how many times is below the required SNR threshold. Moreover, the achieved throughput of each device can be continuously monitored and recorded by the BS. The required threshold and transmission powers can either be pre-configured or defined during the BS-device association phase.

VI. CONCLUSION

In this paper, we introduced a novel SGF scheme that ensures connectivity for two device classes, namely the PDs and SDs. In more detail, by accounting for the sporadic transmission of the SDs, we reported a priority-based SFG NOMA scheme, and we presented a theoretical framework that quantifies its performance in terms of OP and throughput. We derived low-complexity and insightful closed-form expressions for the respective OP and throughput of each device, which are expected to serve as cornerstones of transmission parameters selection. The results revealed the capabilities of the presented scheme to triple the network's capacity while satisfying the SNR requirements.

APPENDIX A

PROOF OF THEOREM 1

The OP of the PD in Case 1 is defined as

$$P_{o,P_m}^{C_1} = \Pr \left[\gamma_{P_m} < \gamma_{th}^{P_m} \right]. \quad (39)$$

By taking into account that in this case $\theta_k = \theta_l = 0$, and employing (10), (39), can be rewritten as

$$P_{o,P_m}^{C_1} = \Pr \left[\frac{E_{P_m} |h_{P_m}|^2}{N_o} < \gamma_{th}^{P_m} \right] \quad (40)$$

or equivalently

$$P_{o,P_m}^{C_1} = \Pr \left[|h_{P_m}|^2 < \frac{\gamma_{th}^{P_m} N_o}{E_{P_m}} \right] \quad (41)$$

or

$$P_{o,P_m}^{C_1} = F_{|h_{P_m}|^2} \left(\frac{\gamma_{th}^{P_m} N_o}{E_{P_m}} \right), \quad (42)$$

where $F_{|h_{P_m}|^2}(\cdot)$ stands for the CDF of $|h_{P_m}|^2$. Note that $|h_{P_m}|$ is a Rayleigh distributed RV with scale parameter equals to 1; thus, $|h_{P_m}|^2$ follows an exponential distribution, with CDF that can be expressed as

$$F_{|h_{P_m}|^2}(x) = 1 - \exp \left(-\frac{x}{2} \right). \quad (43)$$

By applying (43) in (42), we obtain (13). This concludes the proof.

APPENDIX B PROOF OF THEOREM 2

The OP of the PD in Case 2 is defined as in (39). In this case $\theta_k = 1 - \theta_l = 1$; thus, after applying (10), the OP can be rewritten as

$$P_{o,P_m}^{C_2} = \Pr \left[\frac{E_{P_m} |h_{P_m}|^2}{E_{S_k} |h_{S_k}|^2 + N_o} < \gamma_{th}^{P_m} \right], \quad (44)$$

which can be equivalently expressed as

$$P_{o,P_m}^{C_2} = \Pr \left[E_{P_m} |h_{P_m}|^2 < (E_{S_k} |h_{S_k}|^2 + N_o) \gamma_{th}^{P_m} \right]. \quad (45)$$

Note that $|h_{P_m}|^2$ and $|h_{S_k}|^2$ are independent RVs. Hence, according to [50], (45) can be evaluated as

$$P_{o,P_m}^{C_2} = \int_0^\infty F_{|h_{P_m}|^2} \left(\sqrt{\frac{\gamma_{th}^{P_m}}{E_{P_m}}} \sqrt{E_{S_k} y^2 + N_o} \right) f_{|h_{S_k}|^2}(y) dy, \quad (46)$$

where

$$F_{|h_{P_m}|^2}(x) = 1 - \exp \left(-\frac{x^2}{2} \right) \quad (47)$$

is the CDF of $|h_{P_m}|$, and $f_{|h_{S_k}|^2}(x)$ represents the PDF of $|h_{S_k}|$. Note that $|h_{S_k}|$ is an ordered Rayleigh distributed RV with scale parameter equals to 1 and $|h_{S_k}| < |h_{S_l}|$; thus, its PDF can be expressed as

$$f_{|h_{S_k}|^2}(x) = 2x \exp(-x^2). \quad (48)$$

By applying (47) and (48) in (46) and after some algebraic manipulations, we obtain

$$P_{o,P_m}^{C_2} = 1 - 2 \exp\left(-\frac{\gamma_{th}^{P_m} N_o}{2E_{P_m}}\right) \mathcal{I}_1, \quad (49)$$

where

$$\mathcal{I}_1 = \int_0^\infty y \exp\left(-\left(\frac{\gamma_{th}^{P_m} E_{S_k}}{2E_{P_m}} + 1\right) y^2\right) dy. \quad (50)$$

With the aid of [51, eq. (3.321/4)], (50) can be evaluated as

$$\mathcal{I}_1 = \frac{1}{\frac{\gamma_{th}^{P_m} E_{S_k}}{E_{P_m}} + 2}. \quad (51)$$

Finally, by applying (51) in (49), we obtain (14).

Assuming that the decoding order in BS is (s_{P_m}, s_{S_k}) , the OP of S_k in case 2, can be defined as

$$P_{o,S_k}^{C_2}(\gamma_{th}^{S_k}) = 1 - \Pr\left(\gamma_{P_m} \geq \gamma_{th}^{P_m}, \gamma_{S_k} \geq \gamma_{th}^{S_k} | \theta_k = 1, \theta_l = 0\right), \quad (52)$$

which, by taking into account (10) and (11), can be rewritten as

$$P_{o,S_k}^{C_2}(\gamma_{th}^{S_k}) = 1 - \Pr\left(\frac{\gamma_{th}^{S_k} N_o}{E_{S_k}} < |h_{S_k}|^2 < |h_{P_m}|^2 \frac{E_{P_m}}{E_{S_k} \gamma_{th}^{P_m}} - \frac{N_o}{E_{S_k}}\right) \quad (53)$$

or equivalently

$$P_{o,S_k}^{C_2}(\gamma_{th}^{S_k}) = 1 - (\mathcal{F}_1 - \mathcal{F}_2) \quad (54)$$

where

$$\mathcal{F}_1 = \Pr\left(|h_{S_k}|^2 < |h_{P_m}|^2 \frac{E_{P_m}}{E_{S_k} \gamma_{th}^{P_m}} - \frac{N_o}{E_{S_k}}\right) \quad (55)$$

and

$$\mathcal{F}_2 = \Pr\left(|h_{S_k}|^2 < \frac{\gamma_{th}^{S_k} N_o}{E_{S_k}}\right). \quad (56)$$

Next, we evaluate $\mathcal{F}_1$ and $\mathcal{F}_2$. To evaluate (55), we exploit the fact that $|h_{S_k}|^2$ and $|h_{P_m}|^2$ are independent RVs; thus, (55) can be calculated as

$$\mathcal{F}_1 = \int_{\frac{N_o \gamma_{th}^{P_m}}{E_{P_m}}}^\infty F_{|h_{S_k}|^2}\left(y^2 \frac{E_{P_m}}{E_{S_k} \gamma_{th}^{P_m}} - \frac{N_o}{E_{S_k}}\right) f_{|h_{P_m}|}(y) dy, \quad (57)$$

where $F_{|h_{S_k}|^2}(\cdot)$ is the CDF of $|h_{S_k}|^2$ that can be obtained as

$$F_{|h_{S_k}|^2}(x) = 1 - \exp(-x), \quad (58)$$

while $f_{|h_{P_m}|}(\cdot)$ is the PDF of $|h_{P_m}|$ and can be expressed as

$$f_{|h_{P_m}|}(x) = x \exp\left(-\frac{x^2}{2}\right). \quad (59)$$

By applying (58) and (59) in (57), we get

$$\mathcal{F}_1 = \int_{\frac{N_o \gamma_{th}^{P_m}}{E_{P_m}}}^\infty \left(1 - \exp\left(-\left(y^2 \frac{E_{P_m}}{E_{S_k} \gamma_{th}^{P_m}} - \frac{N_o}{E_{S_k}}\right)\right)\right) \times y \exp\left(-\frac{y^2}{2}\right) dy, \quad (60)$$

or equivalently

$$\mathcal{F}_1 = \mathcal{F}_{11} - \mathcal{F}_{12}, \quad (61)$$

where

$$\mathcal{F}_{11} = \int_{\frac{N_o \gamma_{th}^{P_m}}{E_{P_m}}}^\infty y \exp\left(-\frac{y^2}{2}\right) dy \quad (62)$$

and

$$\mathcal{F}_{12} = \int_{\frac{N_o \gamma_{th}^{P_m}}{E_{P_m}}}^\infty y \exp\left(-\left(y^2 \frac{E_{P_m}}{E_{S_k} \gamma_{th}^{P_m}} + \frac{y^2}{2} - \frac{N_o}{E_{S_k}}\right)\right) dy \quad (63)$$

By applying [51, eq. (3.321/4)] in (62), we obtain

$$\mathcal{F}_{11} = \exp\left(-\frac{1}{2} \left(\frac{N_o \gamma_{th}^{P_m}}{E_{P_m}}\right)^2\right). \quad (64)$$

Likewise, after some algebraic manipulations, (63) can be rewritten as

$$\mathcal{F}_{12} = \exp\left(\frac{N_o}{E_{S_k}}\right) \times \int_{\frac{N_o \gamma_{th}^{P_m}}{E_{P_m}}}^\infty y \exp\left(-\left(\frac{E_{P_m}}{E_{S_k} \gamma_{th}^{P_m}} + \frac{1}{2}\right) y^2\right) dy. \quad (65)$$

By applying [51, eq. (3.321/4)] in (65), we obtain

$$\mathcal{F}_{12} = \frac{\exp\left(\frac{N_o}{E_{S_k}} - \left(\frac{N_o \gamma_{th}^{P_m}}{E_{P_m}}\right)^2 \left(\frac{E_{P_m}}{2E_{S_k} \gamma_{th}^{P_m}} + \frac{1}{4}\right)\right)}{\frac{E_{P_m}}{E_{S_k} \gamma_{th}^{P_m}} + \frac{1}{2}} \quad (66)$$

By employing (64) and (66), (61) can be expressed as

$$\mathcal{F}_1 = \exp\left(-\frac{1}{2} \left(\frac{N_o \gamma_{th}^{P_m}}{E_{P_m}}\right)^2\right) - \frac{\exp\left(\frac{N_o}{E_{S_k}} - \left(\frac{N_o \gamma_{th}^{P_m}}{E_{P_m}}\right)^2 \left(\frac{E_{P_m}}{2E_{S_k} \gamma_{th}^{P_m}} + \frac{1}{4}\right)\right)}{\frac{E_{P_m}}{E_{S_k} \gamma_{th}^{P_m}} + \frac{1}{2}}. \quad (67)$$

Next, we evaluate $\mathcal{F}_2$. From (56), we can rewrite $\mathcal{F}_2$ as

$$\mathcal{F}_2 = F_{|h_{S_k}|^2}\left(\frac{\gamma_{th}^{S_k} N_o}{E_{S_k}}\right), \quad (68)$$

which, by applying (58), concludes to

$$\mathcal{F}_2 = 1 - \exp\left(-\frac{\gamma_{th}^{S_k} N_o}{E_{S_k}}\right). \quad (69)$$

Finally, by applying (67) and (69) in (54), we obtain

$$P_{o,S_k}^{C_2}(\gamma_{th}^{S_k}) = 1 - \left(\exp \left(-\frac{1}{2} \left(\frac{N_o \gamma_{th}^{P_m}}{E_{P_m}} \right)^2 \right) \exp \left(\frac{N_o}{E_{S_k}} - \left(\frac{N_o \gamma_{th}^{P_m}}{E_{P_m}} \right)^2 \left(\frac{E_{P_m}}{2E_{S_k} \gamma_{th}^{P_m}} + \frac{1}{4} \right) \right) - \frac{E_{P_m}}{E_{S_k} \gamma_{th}^{P_m}} + \frac{1}{2} \right) - 1 + \exp \left(-\frac{\gamma_{th}^{S_k} N_o}{E_{S_k}} \right) \quad (70)$$

After some algebraic manipulations (70) can be rewritten as (15). This concludes the proof.

APPENDIX C PROOF OF THEOREM 3

The OP of the PD in Case 3 is defined as in (39), assuming that $\theta_k = 0$ and $\theta_l = 1$. By applying (10) in (39) for $\theta_k = 0$ and $\theta_l = 1$, we can rewrite the OP of the PD, in this case, as

$$P_{o,P_m}^{C_3} = \Pr \left[\frac{E_{P_m} |h_{P_m}|^2}{E_{S_l} |h_{S_l}|^2 + N_o} < \gamma_{th}^{P_m} \right] \quad (71)$$

or equivalently

$$P_{o,P_m}^{C_3} = \Pr \left[|h_{P_m}|^2 < \gamma_{th}^{P_m} \frac{E_{S_l}}{E_{P_m}} |h_{S_l}|^2 + \gamma_{th}^{P_m} \frac{N_o}{E_{P_m}} \right]. \quad (72)$$

Notice that $|h_{P_m}|$ and $|h_{S_l}|$ are independent RVs; thus, (72) can be evaluated as

$$P_{o,P_m}^{C_3} = \int_0^\infty \left[F_{|h_{P_m}|} \left(\sqrt{\gamma_{th}^{P_m} \left(\frac{E_{S_l}}{E_{P_m}} y^2 + \frac{N_o}{E_{P_m}} \right)} \right) \times f_{|h_{S_l}|}(y) dy \right], \quad (73)$$

where $f_{|h_{S_l}|}(\cdot)$ represents the PDF of $|h_{S_l}|$. Note that $|h_{S_l}|$ is an ordered Rayleigh distributed RV with scale parameter that is equal to 1 and $|h_{S_k}| < |h_{S_l}|$; hence, the PDF of $|h_{S_l}|$ can be written as

$$f_{|h_{S_l}|}(x) = 2x \left(\exp \left(-\frac{x^2}{2} \right) - \exp(-x^2) \right). \quad (74)$$

By applying (47) and (74) in (73), we obtain

$$P_{o,P_m}^{C_3} = 1 - \mathcal{I}_2, \quad (75)$$

where

$$\mathcal{I}_2 = 2 \int_0^\infty \left[y \exp \left(-\gamma_{th}^{P_m} \left(\frac{E_{S_l}}{E_{P_m}} y^2 + \frac{N_o}{E_{P_m}} \right) \right) \times \left(\exp \left(-\frac{y^2}{2} \right) - \exp(-y^2) \right) dy \right], \quad (76)$$

which can be equivalently written as

$$\mathcal{I}_2 = 2 \exp \left(-\gamma_{th}^{P_m} \frac{N_o}{E_{P_m}} \right) \int_0^\infty \left[y \exp \left(-\gamma_{th}^{P_m} \frac{E_{S_l}}{E_{P_m}} y^2 \right) \times \left(\exp \left(-\frac{y^2}{2} \right) - \exp(-y^2) \right) dy \right], \quad (77)$$

or

$$\mathcal{I}_2 = 2 \exp \left(-\gamma_{th}^{P_m} \frac{N_o}{E_{P_m}} \right) (\mathcal{I}_{21} - \mathcal{I}_{22}), \quad (78)$$

with

$$\mathcal{I}_{21} = \int_0^\infty y \exp \left(-\left(\gamma_{th}^{P_m} \frac{E_{S_l}}{E_{P_m}} + \frac{1}{2} \right) y^2 \right) dy \quad (79)$$

and

$$\mathcal{I}_{22} = \int_0^\infty y \exp \left(-\left(\gamma_{th}^{P_m} \frac{E_{S_l}}{E_{P_m}} + 1 \right) y^2 \right) dy. \quad (80)$$

By applying [51, eq. (3.321/4)] in (79) and (80), we obtain

$$\mathcal{I}_{21} = \frac{1}{2\gamma_{th}^{P_m} \frac{E_{S_l}}{E_{P_m}} + 1} \quad (81)$$

and

$$\mathcal{I}_{22} = \frac{1}{2\gamma_{th}^{P_m} \frac{E_{S_l}}{E_{P_m}} + 2}. \quad (82)$$

Next by employing (81) and (82) in (78), we get

$$\mathcal{I}_2 = 2 \exp \left(-\gamma_{th}^{P_m} \frac{N_o}{E_{P_m}} \right) \times \left(\frac{1}{2\gamma_{th}^{P_m} \frac{E_{S_l}}{E_{P_m}} + 1} - \frac{1}{2\gamma_{th}^{P_m} \frac{E_{S_l}}{E_{P_m}} + 2} \right) \quad (83)$$

or, after some algebraic manipulations,

$$\mathcal{I}_2 = \frac{E_{P_m}^2 \exp \left(-\gamma_{th}^{P_m} \frac{N_o}{E_{P_m}} \right)}{\left(E_{P_m} + E_{S_l} \gamma_{th}^{P_m} \right) \left(E_{P_m} + 2E_{S_l} \gamma_{th}^{P_m} \right)} \quad (84)$$

Finally, by applying (84) in (75), we obtain (16). Assuming that, in this case, the decoding order in BS is (s_{P_m}, s_{S_l}) , the OP of S_k in case 2, can be defined as

$$P_{o,S_l}^{C_3}(\gamma_{th}^{S_l}) = 1 - \Pr \left(\gamma_{P_m} \geq \gamma_{th}^{P_m}, \gamma_{S_l} \geq \gamma_{th}^{S_l} | \theta_k = 0, \theta_l = 1 \right), \quad (85)$$

By taking into account (10) and (12), (85) can be rewritten as

$$P_{o,S_l}^{C_3}(\gamma_{th}^{S_l}) = 1 - \Pr \left(\frac{|h_{P_m}|^2 E_{P_m}}{|h_{S_l}|^2 E_{S_l} + N_o} \geq \gamma_{th}^{P_m}, \frac{|h_{S_l}|^2 E_{S_l}}{N_o} \geq \gamma_{th}^{S_l} \right), \quad (86)$$

which, after some algebraic manipulations, can be expressed as

$$P_{o,S_l}^{C_3}(\gamma_{th}^{S_l}) = 1 - \Pr \left(\gamma_{th}^{S_l} \frac{N_o}{E_{S_l}} \leq |h_{S_l}|^2 \leq \frac{|h_{P_m}|^2 E_{P_m} - \gamma_{th}^{P_m} N_o}{\gamma_{th}^{P_m} E_{S_l}} \right), \quad (87)$$

or

$$P_{o,S_l}^{C_3}(\gamma_{th}^{S_l}) = 1 - (\mathcal{F}_3 - \mathcal{F}_4), \quad (88)$$

where

$$\mathcal{F}_3 = \Pr \left(|h_{S_l}|^2 < \frac{|h_{P_m}|^2 E_{P_m} - \gamma_{th}^{P_m} N_o}{\gamma_{th}^{P_m} E_{S_l}} \right) \quad (89)$$

and

$$\mathcal{F}_4 = \Pr \left(|h_{S_l}|^2 < \gamma_{th}^{S_l} \frac{N_o}{E_{S_l}} \right). \quad (90)$$

Next, we calculate $\mathcal{F}_3$ and $\mathcal{F}_4$. By taking into account that $|h_{S_l}|$ and $|h_{P_m}|$ are independent RVs, (89) can be rewritten as

$$\mathcal{F}_3 = \int_{\frac{N_o \gamma_{th}^{P_m}}{E_{P_m}}}^{\infty} F_{|h_{S_l}|^2} \left(\frac{y^2 E_{P_m} - \gamma_{th}^{P_m} N_o}{\gamma_{th}^{P_m} E_{S_l}} \right) f_{|h_{P_m}|}(y) dy, \quad (91)$$

where $F_{|h_{S_l}|^2}(\cdot)$ is the CDF of $|h_{S_l}|^2$ that can be obtained as

$$F_{|h_{S_l}|^2}(x) = \left(1 - \exp \left(-\frac{x}{2} \right) \right)^2 \quad (92)$$

or equivalently

$$F_{|h_{S_l}|^2}(x) = 1 + \exp(-x) - 2 \exp \left(-\frac{x}{2} \right). \quad (93)$$

and $f_{|h_{P_m}|}(\cdot)$ is the PDF of $|h_{P_m}|$ that can be expressed as

$$f_{|h_{P_m}|}(x) = x \exp \left(-\frac{x^2}{2} \right). \quad (94)$$

By applying (93) and (94) in (91), we obtain

$$\begin{aligned} \mathcal{F}_3 = \int_{\frac{N_o \gamma_{th}^{P_m}}{E_{P_m}}}^{\infty} & \left[\left(1 + \exp \left(-\left(\frac{y^2 E_{P_m} - \gamma_{th}^{P_m} N_o}{\gamma_{th}^{P_m} E_{S_l}} \right) \right) \right. \right. \\ & \left. \left. - 2 \exp \left(-\frac{1}{2} \left(\frac{y^2 E_{P_m} - \gamma_{th}^{P_m} N_o}{\gamma_{th}^{P_m} E_{S_l}} \right) \right) \right) \right. \\ & \left. \times y \exp \left(-\frac{y^2}{2} \right) dy \right] \end{aligned} \quad (95)$$

After some algebraic manipulations (95) can be rewritten as

$$\mathcal{F}_3 = \mathcal{F}_{31} + \mathcal{F}_{32} - 2\mathcal{F}_{33}, \quad (96)$$

where

$$\mathcal{F}_{31} = \int_{\frac{N_o \gamma_{th}^{P_m}}{E_{P_m}}}^{\infty} y \exp \left(-\frac{y^2}{2} \right) dy, \quad (97)$$

$$\begin{aligned} \mathcal{F}_{32} = \exp \left(\frac{N_o}{E_{S_l}} \right) \\ \times \int_{\frac{N_o \gamma_{th}^{P_m}}{E_{P_m}}}^{\infty} y \exp \left(-\left(\frac{E_{P_m}}{\gamma_{th}^{P_m} E_{S_l}} + \frac{1}{2} \right) y^2 \right) dy, \end{aligned} \quad (98)$$

and

$$\begin{aligned} \mathcal{F}_{33} = \exp \left(\frac{N_o}{2E_{S_l}} \right) \\ \times \int_{\frac{N_o \gamma_{th}^{P_m}}{E_{P_m}}}^{\infty} y \exp \left(-\left(\frac{E_{P_m}}{2\gamma_{th}^{P_m} E_{S_l}} + \frac{1}{2} \right) y^2 \right) dy. \end{aligned} \quad (99)$$

By applying [51, eq. (3.321/4)], in (97), (98), and in (99), we respectively get

$$\mathcal{F}_{31} = \exp \left(-\frac{1}{2} \left(\frac{N_o \gamma_{th}^{P_m}}{E_{P_m}} \right)^2 \right), \quad (100)$$

$$\mathcal{F}_{32} = \frac{\exp \left(\frac{N_o}{E_{S_l}} - \frac{1}{2} \left(\frac{N_o \gamma_{th}^{P_m}}{E_{P_m}} \right)^2 \left(\frac{E_{P_m}}{\gamma_{th}^{P_m} E_{S_l}} + \frac{1}{2} \right) \right)}{\frac{E_{P_m}}{\gamma_{th}^{P_m} E_{S_l}} + \frac{1}{2}}, \quad (101)$$

and

$$\mathcal{F}_{33} = \frac{\exp \left(\frac{N_o}{2E_{S_l}} - \frac{1}{2} \left(\frac{N_o \gamma_{th}^{P_m}}{E_{P_m}} \right)^2 \left(\frac{E_{P_m}}{2\gamma_{th}^{P_m} E_{S_l}} + \frac{1}{2} \right) \right)}{\frac{E_{P_m}}{2\gamma_{th}^{P_m} E_{S_l}} + \frac{1}{2}}. \quad (102)$$

With the aid of (100)–(102), (96) can be written in a closed-form as

$$\begin{aligned} \mathcal{F}_3 = \exp \left(-\frac{1}{2} \left(\frac{N_o \gamma_{th}^{P_m}}{E_{P_m}} \right)^2 \right) \\ + \frac{\exp \left(\frac{N_o}{E_{S_l}} - \frac{1}{2} \left(\frac{N_o \gamma_{th}^{P_m}}{E_{P_m}} \right)^2 \left(\frac{E_{P_m}}{\gamma_{th}^{P_m} E_{S_l}} + \frac{1}{2} \right) \right)}{\frac{E_{P_m}}{\gamma_{th}^{P_m} E_{S_l}} + \frac{1}{2}} \\ - 2 \frac{\exp \left(\frac{N_o}{2E_{S_l}} - \frac{1}{2} \left(\frac{N_o \gamma_{th}^{P_m}}{E_{P_m}} \right)^2 \left(\frac{E_{P_m}}{2\gamma_{th}^{P_m} E_{S_l}} + \frac{1}{2} \right) \right)}{\frac{E_{P_m}}{2\gamma_{th}^{P_m} E_{S_l}} + \frac{1}{2}} \end{aligned} \quad (103)$$

Next, we evaluate $\mathcal{F}_4$. From (90), $\mathcal{F}_4$ can be rewritten as

$$\mathcal{F}_4 = F_{|h_{S_l}|^2} \left(\gamma_{th}^{S_l} \frac{N_o}{E_{S_l}} \right), \quad (104)$$

which, based on (93), can be expressed in closed-form as

$$\mathcal{F}_4 = \left(1 - \exp \left(\gamma_{th}^{S_l} \frac{N_o}{2E_{S_l}} \right) \right)^2. \quad (105)$$

By applying (103) and (105) in (88), we obtain

$$\begin{aligned} P_{o,S_l}^{C_3}(\gamma_{th}^{S_l}) = 1 - \left[\exp \left(-\frac{1}{2} \left(\frac{N_o \gamma_{th}^{P_m}}{E_{P_m}} \right)^2 \right) \right. \\ + \frac{\exp \left(\frac{N_o}{E_{S_l}} - \frac{1}{2} \left(\frac{N_o \gamma_{th}^{P_m}}{E_{P_m}} \right)^2 \left(\frac{E_{P_m}}{\gamma_{th}^{P_m} E_{S_l}} + \frac{1}{2} \right) \right)}{\frac{E_{P_m}}{\gamma_{th}^{P_m} E_{S_l}} + \frac{1}{2}} \\ - 2 \frac{\exp \left(\frac{N_o}{2E_{S_l}} - \frac{1}{2} \left(\frac{N_o \gamma_{th}^{P_m}}{E_{P_m}} \right)^2 \left(\frac{E_{P_m}}{2\gamma_{th}^{P_m} E_{S_l}} + \frac{1}{2} \right) \right)}{\frac{E_{P_m}}{2\gamma_{th}^{P_m} E_{S_l}} + \frac{1}{2}} \\ \left. - \left(1 - \exp \left(\gamma_{th}^{S_l} \frac{N_o}{2E_{S_l}} \right) \right)^2 \right] \end{aligned} \quad (106)$$

After some algebraic manipulations, (106) returns (17). This concludes the proof.

APPENDIX D PROOF OF THEOREM 4

The OP of the PD in Case 4 is defined as in (39), which by employing (10), for $\theta_k = \theta_l = 1$, can be expressed as

$$P_{o,P_m}^{C_4} = \Pr \left[\frac{E_{P_m} |h_{P_m}|^2}{I_{k,l} + N_o} < \gamma_{th}^{P_m} \right], \quad (107)$$

where

$$I_{k,l} = E_{S_k} |h_{S_k}|^2 + E_{S_l} |h_{S_l}|^2. \quad (108)$$

In order to evaluate the OP of the PD, we need first to characterize the distribution of $I_{k,l}$. By taking into account that $|h_{S_k}|^2$ and $|h_{S_l}|^2$ are independent RVs, the CDF of $I_{k,l}$ can be calculated as

$$F_{I_{kl}}(x) = \int_0^{\sqrt{\frac{x}{E_{S_l}}}} F_{|h_{S_k}|} \left(\sqrt{\frac{1}{E_{S_k}}} (x - y^2 E_{S_l}) \right) f_{|h_{S_l}|}(y) dy, \quad (109)$$

which, by applying (47) and (74), and after some algebraic manipulations, can be as

$$F_{I_{kl}}(x) = F_{|h_{S_l}|} \left(\sqrt{\frac{x}{E_{S_l}}} \right) - \int_0^{\sqrt{\frac{x}{E_{S_l}}}} \exp \left(-\frac{x - y^2 E_{S_l}}{E_{S_k}} \right) f_{|h_{S_l}|}(y) dy, \quad (110)$$

or

$$F_{I_{kl}}(x) = \left(1 - \exp \left(1 - \exp \left(-\frac{x}{2E_{S_l}} \right)^2 \right) \right)^2 - 2 \exp \left(-\frac{x}{E_{S_k}} \right) \int_0^{\sqrt{\frac{x}{E_{S_l}}}} y \exp \left(\frac{E_{S_l}}{E_{S_k}} y^2 \right) \times \left(\exp \left(-\frac{y^2}{2} \right) - \exp(-y^2) \right) dy, \quad (111)$$

which, after some algebraic manipulations, and by employing [51, eq. (3.321/4)], we obtain

$$F_{I_{kl}}(x) = 1 - \left(\frac{2E_{S_k}}{E_{S_k} - 2E_{S_l}} - \frac{E_{S_k}}{E_{S_k} - E_{S_l}} \right) \exp \left(-\frac{x}{E_{S_k}} \right) - \frac{E_{S_l}}{E_{S_k} - E_{S_l}} \exp \left(-\frac{x}{E_{S_l}} \right) + \frac{4E_{S_l}}{E_{S_k} - 2E_{S_l}} \exp \left(-\frac{x}{2E_{S_l}} \right) \quad (112)$$

Moreover, the PDF of I_{kl} can be obtained as

$$f_{I_{kl}}(x) = \frac{dF_{I_{kl}}(x)}{dx} \quad (113)$$

or

$$f_{I_{kl}}(x) = \left(\frac{2}{E_{S_k} - 2E_{S_l}} - \frac{1}{E_{S_k} - E_{S_l}} \right) \exp \left(-\frac{x}{E_{S_k}} \right) + \frac{1}{E_{S_k} - E_{S_l}} \exp \left(-\frac{x}{E_{S_l}} \right) - \frac{2}{E_{S_k} - 2E_{S_l}} \exp \left(-\frac{x}{2E_{S_l}} \right) \quad (114)$$

By accounting the fact that $|h_{P_m}|$ and I_{kl} are independent RV, (107) can be evaluated as

$$P_{o,P_m}^{C_4} = \int_0^\infty F_{|h_{P_m}|} \left(\sqrt{\frac{\gamma_{th}^{P_m}}{E_{P_m}}} \sqrt{x+1} \right) f_{I_{kl}}(x) dx, \quad (115)$$

or, after employing (47) and (114), returns

$$P_{o,P_m}^{C_4} = 1 - \exp \left(-\frac{\gamma_{th}^{P_m}}{2E_{P_m}} \right) \left(\frac{2}{E_{S_k} - 2E_{S_l}} - \frac{1}{E_{S_k} - E_{S_l}} \right) \times \int_0^\infty \exp \left(-\frac{x}{E_{S_k}} - \frac{\gamma_{th}^{P_m}}{2E_{P_m}} x \right) dx - \frac{\exp \left(-\frac{\gamma_{th}^{P_m}}{2E_{P_m}} \right)}{E_{S_k} - E_{S_l}} \int_0^\infty \exp \left(-\frac{x}{E_{S_l}} - \frac{\gamma_{th}^{P_m}}{2E_{P_m}} x \right) dx + \frac{2 \exp \left(-\frac{\gamma_{th}^{P_m}}{2E_{P_m}} \right)}{E_{S_k} - 2E_{S_l}} \int_0^\infty \exp \left(-\frac{x}{2E_{S_l}} - \frac{\gamma_{th}^{P_m}}{2E_{P_m}} x \right) dx \quad (116)$$

By applying [51, eq. (3.321/3)] in (116) and after some algebraic manipulations, we obtain (18). Next, we need to evaluate the OP of S_k in Case 4, which is defined as

$$P_{o,S_k}^{C_4} = 1 - \Pr \left(\gamma_{P_m} \geq \gamma_{th}^{P_m}, \gamma_{S_k} \geq \gamma_{th}^{S_k} | \theta_k = \theta_l = 1 \right), \quad (117)$$

which, with the aid of (10) and (11), can be equivalently expressed as

$$P_{o,S_k}^{C_4} = 1 - \Pr \left(\frac{\gamma_{th}^{S_k} (E_{S_l} |h_{S_l}|^2 + N_o)}{E_{P_m}} \leq |h_{S_k}|^2 \leq \frac{\mathcal{A}}{E_{S_k} \gamma_{th}^{P_m}} \right), \quad (118)$$

where

$$\mathcal{A} = E_{P_m} |h_{P_m}|^2 - \gamma_{th}^{P_m} E_{S_l} |h_{S_l}|^2 \gamma_{th}^{P_m} N_o. \quad (119)$$

Likewise, (118) can be rewritten as

$$P_{o,S_k}^{C_4} = 1 - (\mathcal{F}_5 - \mathcal{F}_6), \quad (120)$$

with

$$\mathcal{F}_5 = F_{|h_{S_k}|^2} \left(\frac{\mathcal{A}}{E_{S_k} \gamma_{th}^{P_m}} \right) \quad (121)$$

and

$$\mathcal{F}_6 = F_{|h_{S_k}|^2} \left(\frac{\gamma_{th}^{S_k} (E_{S_l} |h_{S_l}|^2 + N_o)}{E_{P_m}} \right). \quad (122)$$

From (120), it becomes evident that in order to evaluate the OP of S_k in Case 4, we need first to derive closed-form expressions for (121) and (122). In this direction, by taking into account that $\mathcal{A}$ and $|h_{S_k}|$ are independent RVs, we can rewrite (121) as

$$\mathcal{F}_5 = \int_0^\infty F_{|h_{S_k}|^2} \left(\frac{y}{E_{S_k} \gamma_{th}^{P_m}} \right) f_{\mathcal{A}}(y) dy, \quad (123)$$

where $f_{\mathcal{A}}(y)$ is the PDF of $\mathcal{A}$. In other words, in order to evaluate $\mathcal{F}_5$, it is necessary to statistically characterize $\mathcal{A}$. From (119), the CDF of $\mathcal{A}$ can be expressed as

$$F_{\mathcal{A}}(x) = \Pr[\mathcal{A} < x], \quad (124)$$

or equivalently

$$F_{\mathcal{A}}(x) = \Pr\left[|h_{P_m}|^2 < \frac{\gamma_{th}^{P_m} E_{S_l} |h_{S_l}|^2 + \gamma_{th}^{P_m} N_o + x}{E_{P_m}}\right]. \quad (125)$$

Notice that $|h_{P_m}|^2$ and $|h_{S_l}|^2$ are independent RVs; thus, (125) can be analytically calculated as

$$F_{\mathcal{A}}(x) = \int_0^\infty \left[F_{|h_{P_m}|^2} \left(\sqrt{\frac{\gamma_{th}^{P_m} E_{S_l} y^2 + \gamma_{th}^{P_m} N_o + x}{E_{P_m}}} \right) \times f_{|h_{S_l}|^2}(y) dy \right]. \quad (126)$$

or, after applying (43) and (74), we obtain

$$F_{\mathcal{A}}(x) = 1 - \int_0^\infty y \exp\left(-\frac{\gamma_{th}^{P_m} E_{S_l} y^2 + \gamma_{th}^{P_m} N_o + x}{2E_{P_m}}\right) \times \left(\exp\left(-\frac{y^2}{2}\right) - \exp(-y^2) \right) dy \quad (127)$$

or equivalently

$$F_{\mathcal{A}}(x) = 1 - \exp\left(-\frac{\gamma_{th}^{P_m} N_o + x}{2E_{P_m}}\right) (\mathcal{J}_1 - \mathcal{J}_2), \quad (128)$$

where

$$\mathcal{J}_1 = \int_0^\infty y \exp\left(-\frac{\gamma_{th}^{P_m} E_{S_l} y^2}{2E_{P_m}} - \frac{y^2}{2}\right) dy \quad (129)$$

and

$$\mathcal{J}_2 = \int_0^\infty y \exp\left(-\frac{\gamma_{th}^{P_m} E_{S_l} y^2}{2E_{P_m}} - y^2\right) dy. \quad (130)$$

By applying [51, eq. (3.321/4)] in (129) and (130), we obtain

$$\mathcal{J}_1 = \frac{2E_{P_m}}{\gamma_{th}^{P_m} E_{S_l} + E_{P_m}} \quad (131)$$

and

$$\mathcal{J}_2 = \frac{2E_{P_m}}{\gamma_{th}^{P_m} E_{S_l} + 2E_{P_m}}. \quad (132)$$

By applying (131) and (132) in (128), we obtain

$$F_{\mathcal{A}}(x) = 1 - \frac{2E_{P_m}^2}{\left(\gamma_{th}^{P_m} E_{S_l} + E_{P_m}\right) \left(\gamma_{th}^{P_m} E_{S_l} + 2E_{P_m}\right)} \times \exp\left(-\frac{\gamma_{th}^{P_m} N_o + x}{2E_{P_m}}\right). \quad (133)$$

The PDF of $\mathcal{A}$ can be obtained as

$$f_{\mathcal{A}}(x) = \frac{dF_{\mathcal{A}}(x)}{dx}, \quad (134)$$

which, after replacing (133) and some basic algebraic manipulations, can be written in closed-form as

$$f_{\mathcal{A}}(x) = \frac{E_{P_m} \exp\left(-\frac{\gamma_{th}^{P_m} N_o + x}{2E_{P_m}}\right)}{\left(\gamma_{th}^{P_m} E_{S_l} + E_{P_m}\right) \left(\gamma_{th}^{P_m} E_{S_l} + 2E_{P_m}\right)}. \quad (135)$$

By applying (47) and (135) in (123)

$$\mathcal{F}_5 = 1 - \frac{E_{P_m} \exp\left(-\frac{\gamma_{th}^{P_m} N_o}{2E_{P_m}}\right)}{\left(\gamma_{th}^{P_m} E_{S_l} + E_{P_m}\right) \left(\gamma_{th}^{P_m} E_{S_l} + 2E_{P_m}\right)} \times \int_0^\infty \exp\left(-\frac{y}{2E_{S_k} \gamma_{th}^{P_m}} - \frac{y}{2E_{P_m}}\right) dy, \quad (136)$$

which can be written in a closed-form as

$$\mathcal{F}_5 = 1 - \frac{E_{P_m} \exp\left(-\frac{\gamma_{th}^{P_m} N_o}{2E_{P_m}}\right)}{\left(\gamma_{th}^{P_m} E_{S_l} + E_{P_m}\right) \left(\gamma_{th}^{P_m} E_{S_l} + 2E_{P_m}\right)} \times \frac{2}{\frac{1}{E_{S_k} \gamma_{th}^{P_m}} + \frac{1}{E_{P_m}}} \quad (137)$$

or as

$$\mathcal{F}_5 = 1 - \frac{2E_{P_m}^2 E_{S_k} \gamma_{th}^{P_m}}{\left(\gamma_{th}^{P_m} E_{S_l} + E_{P_m}\right) \left(\gamma_{th}^{P_m} E_{S_k} + E_{P_m}\right)} \times \frac{\exp\left(-\frac{\gamma_{th}^{P_m} N_o}{2E_{P_m}}\right)}{\left(\gamma_{th}^{P_m} E_{S_l} + 2E_{P_m}\right)}. \quad (138)$$

Similarly, (122) can be evaluated as

$$\mathcal{F}_6 = \int_0^\infty F_{|h_{S_k}|^2} \left(\frac{\gamma_{th}^{S_k} (E_{S_l} y^2 + N_o)}{E_{P_m}} \right) f_{|h_{S_l}|^2}(y) dy \quad (139)$$

or, after applying (58) and (74) and conducting basic algebraic manipulations, returns

$$\mathcal{F}_6 = 1 + \frac{E_{S_k} \exp\left(-\frac{\gamma_{th}^{S_k}}{E_{S_k}}\right)}{\gamma_{th}^{S_k} E_{S_l} + E_{S_k}} - \frac{4E_{S_k} \exp\left(-\frac{\gamma_{th}^{S_k}}{2E_{S_k}}\right)}{\gamma_{th}^{S_k} E_{S_l} + 2E_{S_k}}. \quad (140)$$

By applying (138) and (140) in (120), we obtain (141) shown at the top of the next page.

Finally, the OP of S_l in Case 4 is defined as

$$P_{o,S_l}^{C_4} \left(\gamma_{th}^{S_l} \right) = 1 - \Pr \left[\gamma_{P_m} \geq \gamma_{th}^{P_m}, \gamma_{S_k} \geq \gamma_{th}^{S_k}, \gamma_{S_l} \geq \gamma_{th}^{S_l} \mid \theta_k = \theta_l = 1 \right], \quad (142)$$

$$P_{o,S_k}^{C_4} = 1 - \left(\frac{2E_{P_m}^2 E_{S_k} \gamma_{th}^{P_m} \exp\left(-\frac{\gamma_{th}^{P_m} N_o}{2E_{P_m}}\right)}{\left(\gamma_{th}^{P_m} E_{S_l} + E_{P_m}\right) \left(\gamma_{th}^{P_m} E_{S_k} + E_{P_m}\right) \left(\gamma_{th}^{P_m} E_{S_l} + 2E_{P_m}\right)} - \frac{E_{S_k} \exp\left(-\frac{\gamma_{th}^{S_k}}{E_{S_k}}\right)}{\gamma_{th}^{S_k} E_{S_l} + E_{S_k}} - \frac{4E_{S_k} \exp\left(-\frac{\gamma_{th}^{S_k}}{2E_{S_k}}\right)}{\gamma_{th}^{S_k} E_{S_l} + 2E_{S_k}} \right) \quad (141)$$

which can be rewritten as

$$P_{o,S_l}^{C_4} \left(\gamma_{th}^{S_l} \right) = 1 - \Pr \left[\frac{N_o \gamma_{th}^{S_l}}{E_{S_l}} < |h_{S_l}|^2 \right. \\ \left. < \min \left(\frac{E_{P_m} |h_{P_m}|^2 - \gamma_{th}^{P_m} E_{S_k} |h_{S_k}|^2 - \gamma_{th}^{P_m} N_o}{\gamma_{th}^{P_m} E_{S_l}}, \right. \right. \\ \left. \left. \frac{E_{S_k} |h_{S_k}|^2 \gamma_{th}^{S_k} - N_o \gamma_{th}^{S_k}}{E_{S_l} \gamma_{th}^{S_k}} \right) \right] \quad (143)$$

By taking into account the basic assumption of this protocol, i.e., $E_{P_m} |h_{P_m}|^2 \gg E_{S_k} |h_{S_k}|^2$, (143) can be simplified as

$$P_{o,S_l}^{C_4} \left(\gamma_{th}^{S_l} \right) \approx 1 \\ - \Pr \left[\frac{N_o \gamma_{th}^{S_l}}{E_{S_l}} < |h_{S_l}|^2 < \frac{E_{S_k} |h_{S_k}|^2 \gamma_{th}^{S_k} - N_o \gamma_{th}^{S_k}}{E_{S_l} \gamma_{th}^{S_k}} \right] \quad (144)$$

or equivalently

$$P_{o,S_l}^{C_4} \left(\gamma_{th}^{S_l} \right) = 1 - (\mathcal{F}_7 - \mathcal{F}_8), \quad (145)$$

where

$$\mathcal{F}_7 = \Pr \left[|h_{S_l}|^2 < \frac{E_{S_k} |h_{S_k}|^2 \gamma_{th}^{S_k} - N_o \gamma_{th}^{S_k}}{E_{S_l} \gamma_{th}^{S_k}} \right] \quad (146)$$

and

$$\mathcal{F}_8 = \Pr \left[|h_{S_l}|^2 < \frac{N_o \gamma_{th}^{S_l}}{E_{S_l}} \right] \quad (147)$$

By accounting the independency of $|h_{S_l}|$ and $|h_{S_k}|^2$, (146) can be evaluated as

$$\mathcal{F}_7 = \int_0^\infty F_{|h_{S_l}|} \left(\sqrt{\frac{E_{S_k} y^2 \gamma_{th}^{S_k} - N_o \gamma_{th}^{S_k}}{E_{S_l} \gamma_{th}^{S_k}}} \right) f_{|h_{S_k}|}(y) dy, \quad (148)$$

which after applying (93) and (48), can be rewritten as

$$\mathcal{F}_7 = 1 - 2 \exp \left(\frac{N_o \gamma_{th}^{S_k}}{E_{S_l} \gamma_{th}^{S_k}} \right) \int_0^\infty \left[x \exp \left(-\frac{E_{S_k} \gamma_{th}^{S_k}}{E_{S_l} \gamma_{th}^{S_k}} y^2 \right) \right. \\ \left. \times \left(\exp \left(-\frac{y^2}{2} \right) - \exp(-y^2) \right) dy \right], \quad (149)$$

or equivalently

$$\mathcal{F}_7 = 1 - 2 \exp \left(\frac{N_o \gamma_{th}^{S_k}}{E_{S_l} \gamma_{th}^{S_k}} \right) (\mathcal{F}_{71} - \mathcal{F}_{72}), \quad (150)$$

where

$$\mathcal{F}_{71} = \int_0^\infty x \exp \left(-\left(\frac{E_{S_k} \gamma_{th}^{S_k}}{E_{S_l} \gamma_{th}^{S_k}} + \frac{1}{2} \right) y^2 \right) dy \quad (151)$$

and

$$\mathcal{F}_{72} = \int_0^\infty x \exp \left(-\left(\frac{E_{S_k} \gamma_{th}^{S_k}}{E_{S_l} \gamma_{th}^{S_k}} + 1 \right) y^2 \right) dy \quad (152)$$

By applying [51, eq. (3.321/4)] in (151) and (152), we obtain

$$\mathcal{F}_{71} = \frac{E_{S_l} \gamma_{th}^{S_k}}{2E_{S_k} \gamma_{th}^{S_k} + E_{S_l} \gamma_{th}^{S_k}} \text{ and } \mathcal{F}_{72} = \frac{1}{2} \frac{E_{S_l} \gamma_{th}^{S_k}}{E_{S_k} \gamma_{th}^{S_k} + E_{S_l} \gamma_{th}^{S_k}} \quad (153)$$

By applying (153) in (150), we get

$$\mathcal{F}_7 = 1 - 2 \left(\frac{E_{S_l} \gamma_{th}^{S_k}}{2E_{S_k} \gamma_{th}^{S_k} + E_{S_l} \gamma_{th}^{S_k}} - \frac{1}{2} \frac{E_{S_l} \gamma_{th}^{S_k}}{E_{S_k} \gamma_{th}^{S_k} + E_{S_l} \gamma_{th}^{S_k}} \right) \\ \times \exp \left(\frac{N_o \gamma_{th}^{S_k}}{E_{S_l} \gamma_{th}^{S_k}} \right). \quad (154)$$

To evaluate (147), we use (93), and we get

$$\mathcal{F}_8 = \left(1 - \frac{N_o \gamma_{th}^{S_l}}{E_{S_l}} \exp \left(-\frac{N_o \gamma_{th}^{S_l}}{E_{S_l}} \right) \right)^2. \quad (155)$$

By applying (154) and (155) in (145), we obtain (22). This concludes the proof.

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